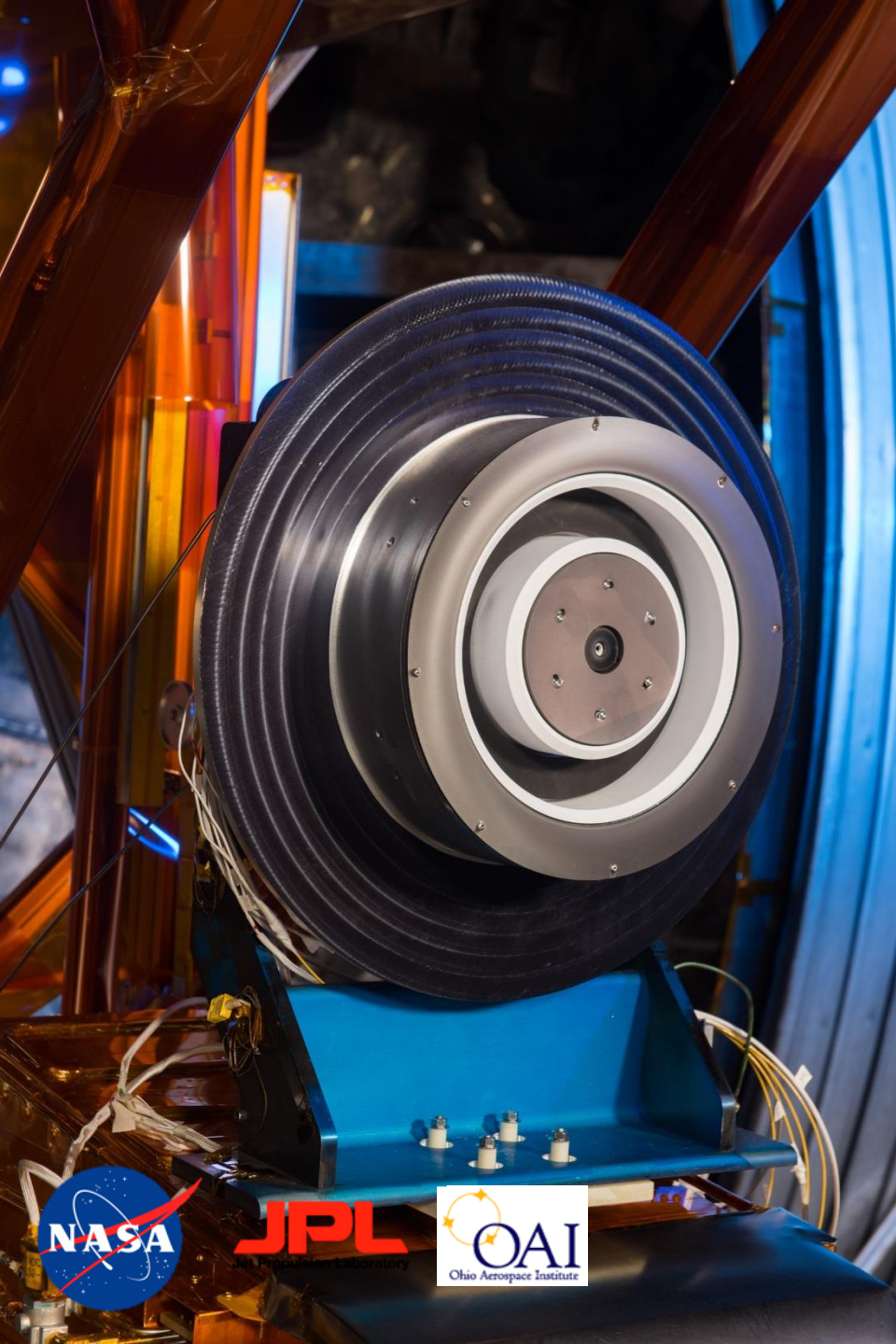




EXTENDED PERFORMANCE CHARACTERIZATION OF THE 12.5- KW ADVANCED ELECTRIC PROPULSION SYSTEM ENGINEERING TEST UNIT HALL THRUSTER

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Introduction

- High-power (40-kW) SEP capability has been identified as enabling for near term and future NASA exploration architectures
 - Example: Maxar Power and Propulsion Element of NASA's Gateway
- Since 2012, NASA has been developing the Hall Effect Rocket with Magnetic Shielding (HERMeS) to serve as a SEP capability building block
- Technology development transitioned to Aerojet Rocketdyne (AR) via Advanced Electric Propulsion System (AEPS) contract
 - AR has produced two AEPS Engineering Test Unit (ETU) thrusters and is leading a detailed test campaign to assess design compliance with AEPS requirements
 - NASA continues to support AEPS development by leveraging in-house expertise, plasma modeling capability, and world-class test facilities





Test Summary

- This work presents a detailed characterization of ETU-2 performance at the reference firing conditions (RFCs) most relevant for AEPS application on PPE
 - Performance also characterized at previously-tested AEPS RFCs to confirm thruster/facility operation matched historical baselines

RFCs	Discharge Voltage (V)	Discharge Current (A)	Discharge Power (W)	Cathode Flow Fraction (%)
AEPS	300	20.83	6250	7
	400	20.83	8333	
	500	20.83	10417	
	600	20.83	12500	
PPE	300	8.6	2600	8
	300	10.0	3000	
	300	15.0	4500	
	300	20.0	6000	
	300	20.83	6250	
	600	15.0	9000	
	600	20.0	12000	
	600	20.0	12000	14



Experimental Apparatus

AR AEPS ETU-2

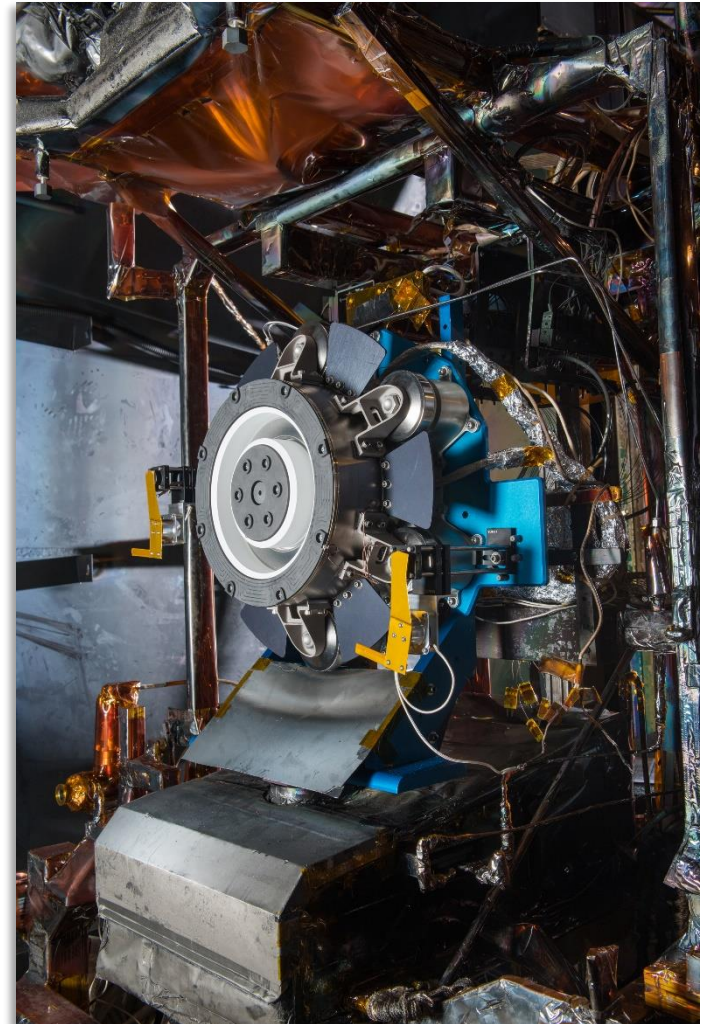
- ETU design derived from and preserves key features of the HERMeS TDUs, but includes modifications to improve manufacturability and ability to meet environmental and spacecraft interface requirements
- Power and propellant supplied using calibrated commercial laboratory systems
 - Flow rate uncertainty: 0.6-1.3 sccm
 - Voltage uncertainty: ± 0.06 V
 - Current uncertainty: ± 0.03 A

GRC VF-5

- Nominal pumping speed: ~ 700 kl/s on xenon
 - Nominal operating pressure: ~ 4 μ Torr at 12.5 kW throttle point

Thrust Stand

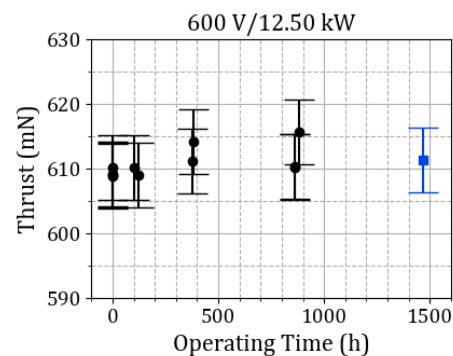
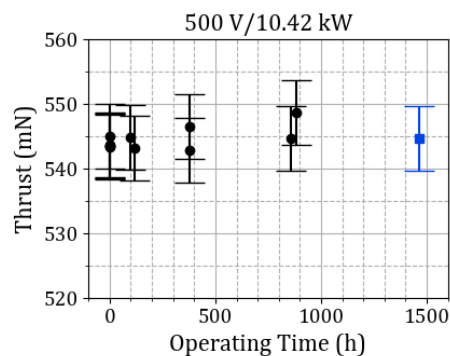
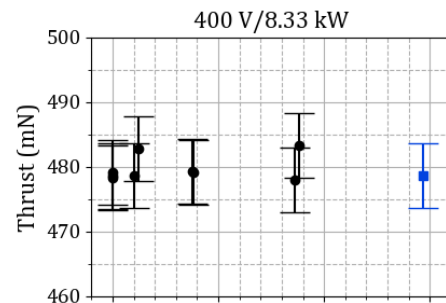
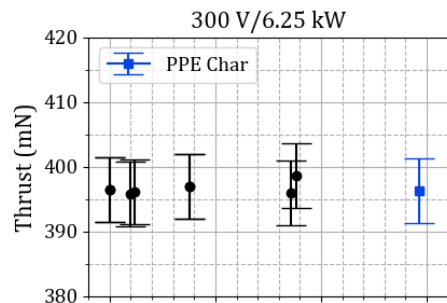
- Thrust measured with an inverted pendulum thrust stand ($\pm 0.8\%$ uncertainty) (AIAA Paper 2018-4516)
- Same thrust stand used during all TDU characterization and wear tests in VF-5





Results: AEPS RFCs

- Thrust of ETU-2 matched historical means to within the empirical uncertainty

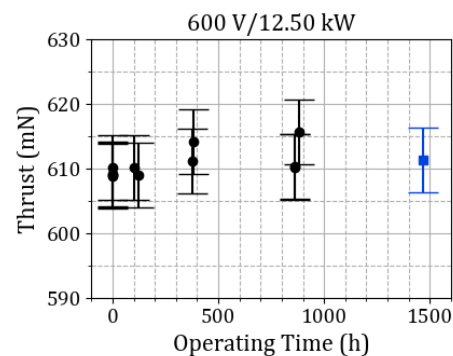
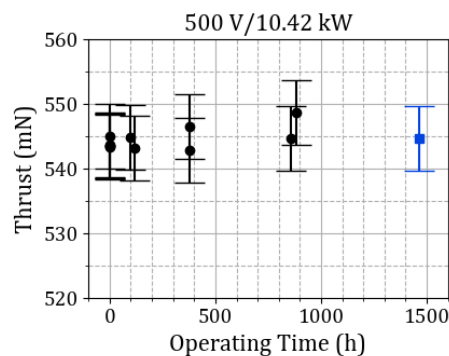
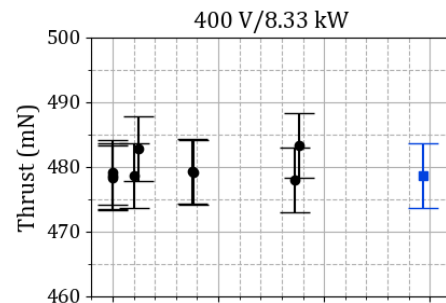
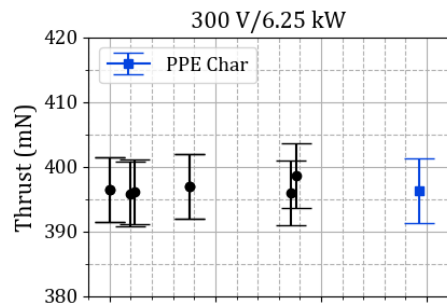


RFC	Thrust (mN)	
	Mean	PPE Char
300 V/6.25 kW	397	396
400 V/8.33 kW	480	479
500 V/10.42 kW	545	545
600 V/12.50 kW	611	611
Uncertainty	± 5 mN	



Results: AEPS RFCs

- Thrust of ETU-2 matched historical means to within the empirical uncertainty
 - Trend held for specific impulse, total efficiency, and discharge current oscillation characteristics

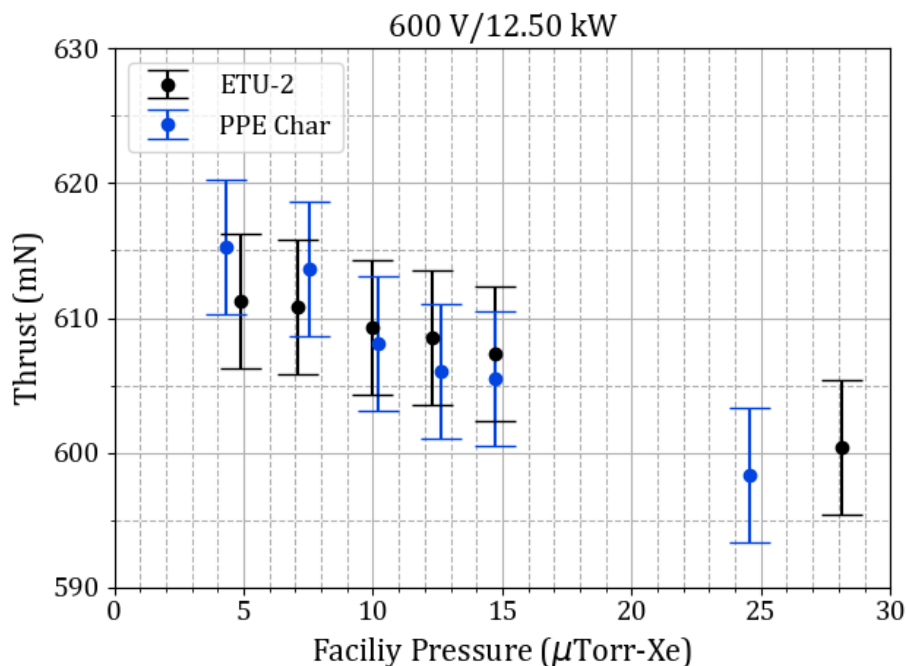


RFC	Thrust (mN)		Specific Impulse (s)		Total Efficiency (%)	
	Mean	PPE Char	Mean	PPE Char	Mean	PPE Char
300 V/6.25 kW	397	396	1960	1972	60.3	60.6
400 V/8.33 kW	480	479	2323	2326	65.0	64.9
500 V/10.42 kW	545	545	2599	2602	66.2	66.2
600 V/12.50 kW	611	611	2817	2830	67.1	69.4
Uncertainty	± 5 mN		± 0.9%		± 1.7%	



Results: AEPS RFCs

- Thrust of ETU-2 matched historical means to within the empirical uncertainty
 - Trend held for specific impulse, total efficiency, and discharge current oscillation characteristics
- Variation in ETU-2 performance with background pressure also matched historical trends
 - Trend held for facility pressure at matched total flow rates equal

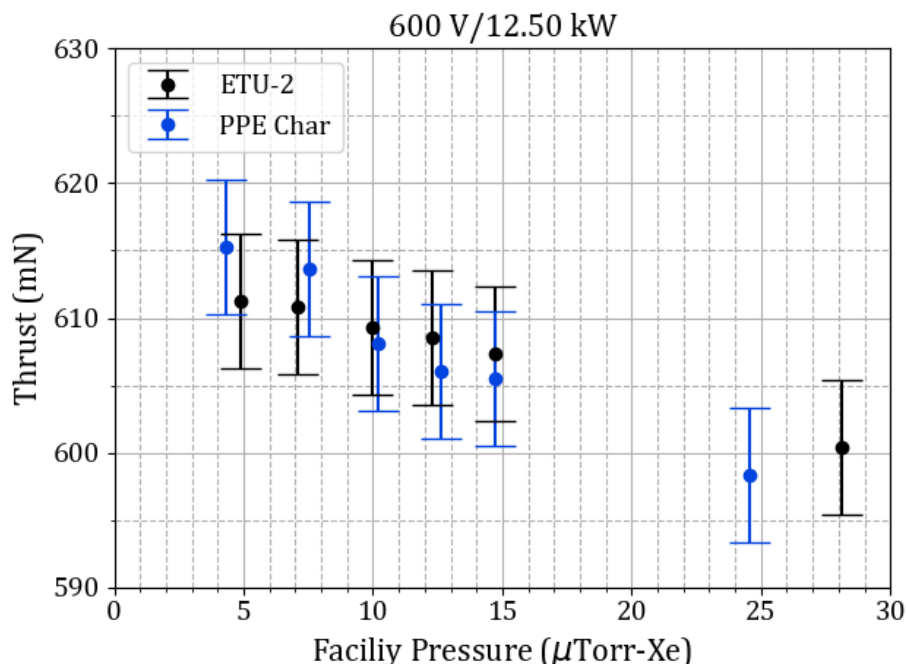


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- Thrust of ETU-2 matched historical means to within the empirical uncertainty
 - Trend held for specific impulse, total efficiency, and discharge current oscillation characteristics
- Variation in ETU-2 performance with background pressure also matched historical trends
 - Trend held for facility pressure at matched total flow rates equal
- Results suggest thruster and facility performance successfully restored following reconfiguration for LIF testing

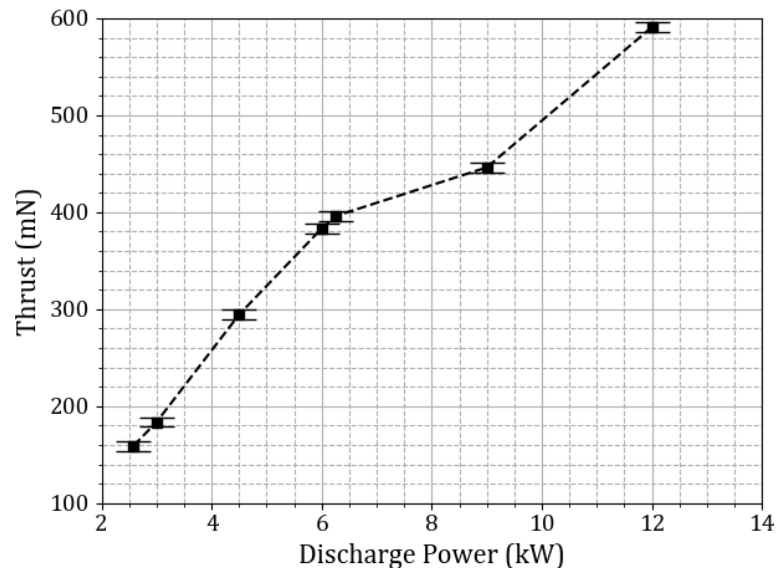


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Results: PPE RFCs

- ETU-2 performance assessed at 8 PPE RFCs
- Thrust of ETU-2 met or exceeded preliminary PPE requirements for all RFCs

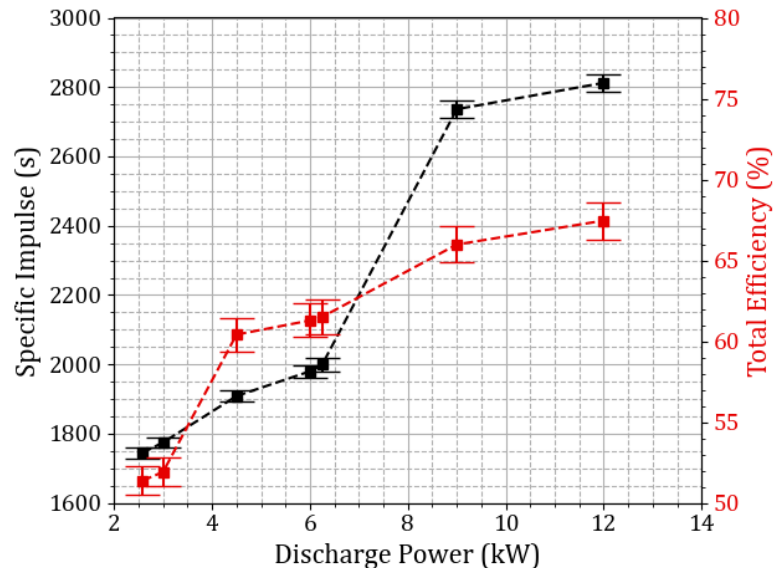


RFC	Thrust (mN)	Specific Impulse (s)	Total Efficiency (%)
300 V/2.60 kW	159	1745	51.4
300 V/3.00 kW	184	1775	51.9
300 V/4.50 kW	294	1909	60.4
300 V/6.00 kW	383	1980	61.3
300 V/6.25 kW	396	1999	61.5
600 V/9.00 kW	446	2736	66.0
600 V/12.00 kW	591	2812	67.5
600 V/12.00 kW/14%	594	2740	66.1
Uncertainty	± 5 mN	± 0.9%	± 1.7%



Results: PPE RFCs

- ETU-2 performance assessed at 8 PPE RFCs
- Thrust of ETU-2 met or exceeded preliminary PPE requirements for all RFCs
 - Trend held for both specific impulse and **total efficiency**

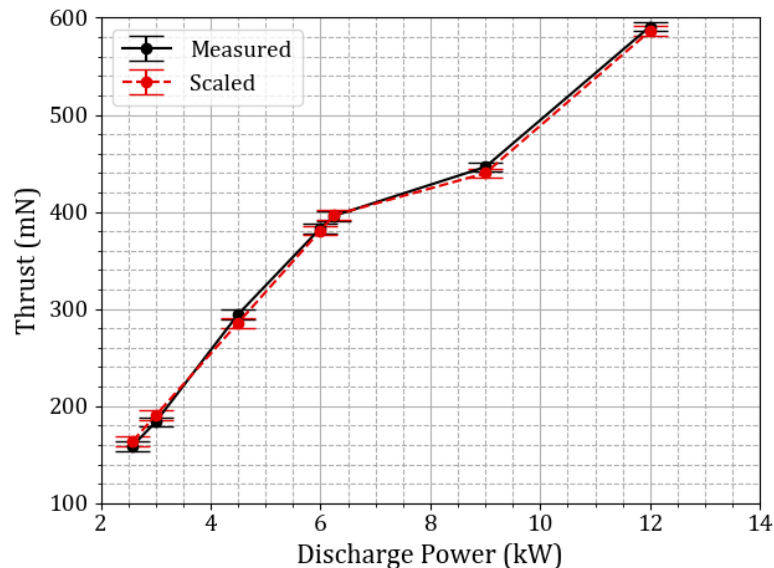


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Results: PPE RFCs

- ETU-2 performance assessed at 8 PPE RFCs
- Thrust of ETU-2 met or exceeded preliminary PPE requirements for all RFCs
 - Trend held for both specific impulse and total efficiency
- Few comparable measurements made on HERMeS TDUs
 - Results are in-family with predictions based on **scaling of previous performance measurements at 300 V and 600 V**



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Conclusions

- A test campaign of the AEPS ETU-2 thruster was performed to assess thruster performance at the throttle conditions most relevant for AEPS application on the Gateway PPE
- ETU-2 performance and backpressure sensitivity were observed to match previous results to within the empirical uncertainty at the nominal AEPS throttle conditions
 - Results confirmed facility and thruster were operating nominally and consistent with previous baselines following facility reconfiguration to accommodate LIF testing
- ETU-2 performance at the expanded set of PPE throttle condition was found to be in-family with predictions based on scaling of prior results
 - Discharge current oscillation characteristics were also observed to match the same general trends observed at the nominal AEPS throttle conditions
- Taken together, these results inform AEPS application on PPE as the design progresses towards its Critical Design Review